PROGRAMMING-FOR-DATA-ANALYTICS-2-PDAN8412-2026

PART 2- ASSIGNMENT

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EXECUTIVE SUMMARY

The book press wanted a Logistic Regression method, which was created from scratch, to predict whether a book would be a best seller. This paper explains how the data was chosen, cleaned, and explored, how the model was created and trained, and how the outcomes were assessed.

The study examined 700 Amazon popular book records (2009-2022). Because every book in the dataset is already a bestseller, a tighter aim known as "top-tier" was established: a book with a User Rating of at least 4.8 and more reviews than the median book. Because User Rating and Reviews define this objective, they were omitted from the predictors to avoid target leakage. The data was also separated by book title, thus there are no titles in either the training or test sets (558 training and 142 test records). Price, Year, and Genre were used as predictors during feature selection.

On the previously unknown test set, the model attained an accuracy of 80.28%, a precision of 0.53, a recall of 0.31, an F1-score of 0.39, and a ROC-AUC of 0.82. The model is competent at recognising standard novels (93% recall), however it only finds three out of ten top-tier books. Its accuracy is only slightly higher than the 79.6% achieved by always predicting "standard," but the ROC-AUC demonstrates that it ranks top-tier books far above chance. It is therefore best used as a screening tool to indicate promising novels, rather than as a final decision-maker.

EVALUATION AND JUSTIFICATION OF THE DATASET

Dataset: a publicly available Kaggle file of Amazon's top-selling books, *bestsellers_with_categories_2022_03_27.csv*. It holds 700 rows and 7 original columns: Name, Author, User Rating, Reviews, Price, Year and Genre.

https://www.kaggle.com/datasets/chriskachmar/amazon-top-50-bestselling-books-2009-2022?resource=download.

Why is it suitable for Logistic Regression classification?

• It contains the algorithm's required data types: numeric predictors (rating, review count, price, year) and a categorical predictor (Genre) that can be represented as 0/1.

• Logistic regression requires a binary outcome. There are no ready-made data sets, but one can be created using rating and review volume (see Section 4).

• It is immediately related to the client's concern because it describes actual books and reader reactions to them.

Data quality

Check	Finding
Missing values	None in any column
Exact duplicate rows	None
Data types	Correct (floats and integers for numbers, text for Name, Author and Genre)
Size	700 rows: small, and only a few independent predictors remain once the label-defining variables are removed
Class balance (after target creation)	545 standard (77.9%) vs 155 top-tier (22.1%): imbalanced

Pitfalls considered:

- There is no natural “best seller” label. Every book is already a bestseller, so a label had to be created from User Rating and Reviews. This is the dataset’s main limitation.
- Target leakage. Because the label is built from User Rating and Reviews, using them as predictors would let the model see the answer (Kaufman et al., 2012). They were excluded from the model.
- Books that appear repeatedly. A book can be listed in several years, so there are only 441 distinct titles across 700 rows. To stop the same title appearing in both the training and test sets, the data was split by title.
- Skewed values. Review counts range from 37 to 208,917 and some prices are extreme (up to $105), so the predictors were scaled.
- A narrow rating range. Ratings range from 3.3 to 4.9, with most books scoring 4.5 or higher.
- Few features. Only a handful of columns exist. Once rating and reviews are removed, only Price, Year, Genre and Title_Length are left, and drivers such as marketing spend or sales volume are not captured.

ANALYSIS PLAN

Stage	Plan
a. Exploratory Data Analysis	Check shape, data types, missing values and duplicates. Review summary statistics. Do data cleaning, target engineering and feature engineering. Generate visualisations: correlation heatmaps, price by genre, rating against reviews, price against reviews, rating distribution, top-tier by genre and by year. Check the class balance of the target (is_top_tier).
b. Feature Selection	Exclude User Rating and Reviews from the predictors because they define the target. Use Price, Year, Title_Length and Genre_Encoded as candidates. Split the data by book title (80/20, random_state = 42) before selection. Fit a statsmodels Logit model on the training set only and keep features with a p-value below 0.05.
c. Train the Model	Standardise the features using the training-set mean and standard deviation only. Train the from-scratch Logistic Regression with batch gradient descent. Planned hyperparameters: learning rate 0.1, 2,000 iterations, weights and bias starting at zero, classification threshold 0.5.
d. Interpret and Evaluate	Evaluate on the unseen test set using accuracy, precision, recall, F1-score, ROC-AUC, the confusion matrix and the ROC curve. Because the classes are imbalanced, prioritise recall, precision and F1-score over accuracy alone.

IMPORT LIBRARIES AND DATASET

To begin the analysis, essential Python libraries for specific data science tasks were imported: numpy (np) for numerical operations, pandas (pd) for structured data handling and manipulation, and matplotlib.pyplot (plt) in conjunction with seaborn (sns) to generate professional exploratory data visualisations. Following library initialisation, random seeding of 42 was set to ensure reproducibility, and the Amazon Bestsellers dataset was imported directly into the workspace into a pandas DataFrame (df) using the pd.read_csv() function from a specified local file directory

(bestsellers_with_categories.csv), successfully loading 700 rows and 7 initial columns for further processing.

a. Exploratory data analysis

DATASET INSPECTION

The dataset assessment revealed that the Amazon Bestsellers dataset has 700 rows and 7 total features, with no missing values or duplicate records, indicating a clean and comprehensive foundation. Numerical descriptive statistics show that user ratings average 4.64 out of 5.0 (median 4.7, min 3.3, max 4.9), review counts average 19,255 (median 10,284, ranging from 37 to 208,917), prices average $12.70 (median $11.00, reaching an outlier max of $105.00), and publication years range from 2009 to 2022. Categorical elements include 441 unique book titles (headed by the APA Publication Manual with 10 appearances), 305 unique authors (led by Jeff Kinney with 14 entries), and two genres, with Nonfiction accounting for 55.4%. The engineered binary target variable (is_top_tier) resulted in a natural 78/22 class imbalance, with 545 standard blockbusters (Class 0, ~77.86%) and 155 elite top-tier bestsellers (Class 1, ~22.14%).

```
============================================================
1. DATASET BASIC INFO & SHAPE
============================================================
Total Rows (Records): 700
Total Columns (Features): 7

Data Types & Non-Null Counts:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 700 entries, 0 to 699
Data columns (total 7 columns):
 #   Column       Non-Null Count  Dtype
---  ------       --------------  -----
 0   Name         700 non-null    object
 1   Author       700 non-null    object
 2   User Rating  700 non-null    float64
 3   Reviews      700 non-null    int64
 4   Price        700 non-null    int64
 5   Year         700 non-null    int64
 6   Genre        700 non-null    object
dtypes: float64(1), int64(3), object(3)
memory usage: 38.4+ KB
None

============================================================
2. FIRST 5 ROWS OF THE DATASET
============================================================
```

	Name	Author	User Rating	Reviews	Price	Year	Genre
0	Act Like a Lady, Think Like a Man: What Men Re...	Steve Harvey	4.6	5013	17	2009	Non Fiction
1	Arguing with Idiots: How to Stop Small Minds a...	Glenn Beck	4.6	798	5	2009	Non Fiction
2	Breaking Dawn (The Twilight Saga, Book 4)	Stephenie Meyer	4.6	9769	13	2009	Fiction
3	Crazy Love: Overwhelmed by a Relentless God	Francis Chan	4.7	1542	14	2009	Non Fiction
4	Dead And Gone: A Sookie Stackhouse Novel (Sook...	Charlaine Harris	4.6	1541	4	2009	Fiction

```
============================================================
3. DESCRIPTIVE STATISTICS (Numerical Features)
============================================================
             count          mean           std     min      25%      50%  \
User Rating  700.0      4.639857      0.218586     3.3     4.50      4.7
Reviews      700.0  19255.195714  23613.443875    37.0  4987.25  10284.0
Price        700.0     12.700000      9.915162     0.0     7.00     11.0
Year         700.0   2015.500000      4.034011  2009.0  2012.00   2015.5

                 75%       max
User Rating      4.8       4.9
Reviews      23358.0  208917.0
Price           15.0     105.0
Year          2019.0    2022.0

============================================================
4. DESCRIPTIVE STATISTICS (Categorical Features)
============================================================
       count unique                                        top freq
Name     700    441  Publication Manual of the American Psychologic...   10
Author   700    305                                  Jeff Kinney   14
Genre    700      2                                  Non Fiction  388
```

```
============================================================
5. UNIQUE VALUES CHECK
============================================================
Unique values in 'Name': 441
Unique values in 'Author': 305
Unique values in 'User Rating': 14
Unique values in 'Reviews': 462
Unique values in 'Price': 40
Unique values in 'Year': 14
Unique values in 'Genre': 2
Missing Values per Column:
 Name           0
Author          0
User Rating     0
Reviews         0
Price           0
Year            0
Genre           0
dtype: int64
Duplicate rows count: 0

--- Target Class Distribution ---
is_top_tier
0    77.857143
1    22.142857
Name: proportion, dtype: float64
is_top_tier
0    545
1    155
Name: count, dtype: int64
```

EDA (DATA CLEANING & TARGET ENGINEERING)

Data cleaning was required to ensure the integrity of the dataset. The checks found zero missing values and zero duplicate rows across all 700 entries, so no rows had to be removed or imputed. A duplicate-removal step was still included in the code in case any duplicates existed.

Target engineering was necessary to convert the raw dataset into a supervised classification task. Every book in the collection is already a historical bestseller, therefore sales rank alone cannot distinguish exceptional blockbusters from mediocre ones. The binary target is_top_tier was thus defined as 1 when the User Rating is at least 4.8 (excellent reader happiness) and the Reviews exceed the median of 10,284 (strong community participation), and 0 otherwise. This yielded 545 standard blockbusters (Class 0, 77.86%) and 155 top-tier bestsellers (Class 1, 22.14%), resulting in a natural 78/22 class imbalance.

Three features were then created: Title_Length (the number of characters in the title), Author_Frequency (how frequently an author occurs in the dataset), and Genre_Encoded (Fiction = 0, Non Fiction = 1). They were checked during feature selection, and Title_Length was not statistically significant (p = 0.283), therefore it

was removed, however Genre_Encoded was significant (p = 0.013) and remained. Author_Frequency was generated but not included as a candidate predictor.

Because User Rating and Reviews define the target, they were kept out of the predictors. This avoids target leakage and makes the final model a fairer test of whether other book details can predict top-tier status.

```
Missing Values per Column:
 Name           0
Author          0
User Rating     0
Reviews         0
Price           0
Year            0
Genre           0
dtype: int64
Duplicate rows count: 0

Target Class Distribution:
is_top_tier
0    545
1    155
Name: count, dtype: int64
```

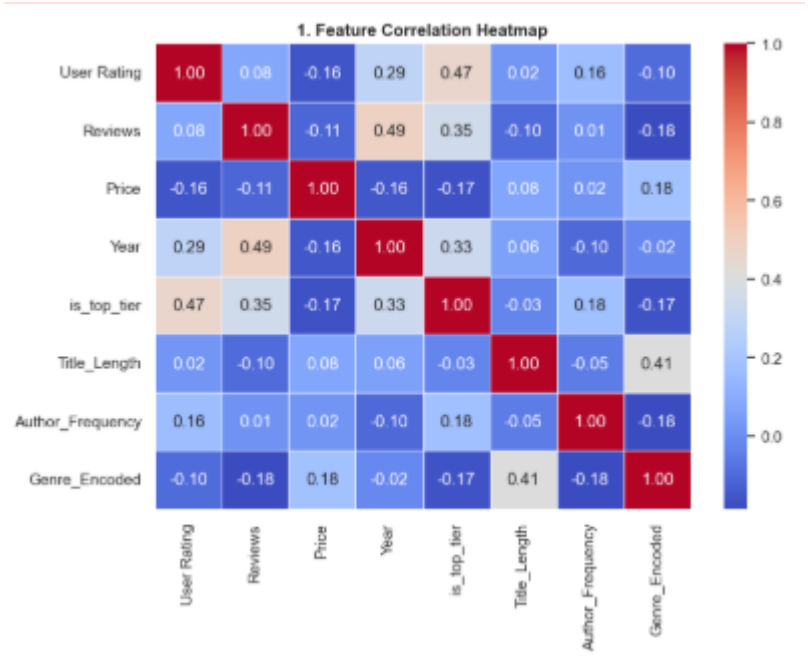
EDA (VISUALISATIONS)

Correlation Heatmap

Top Drivers of Elite Status: The biggest positive association with is_top_tier is seen in User Rating (+0.47) and Reviews (+0.35), indicating that elite bestseller status is driven by outstanding ratings and significant engagement.

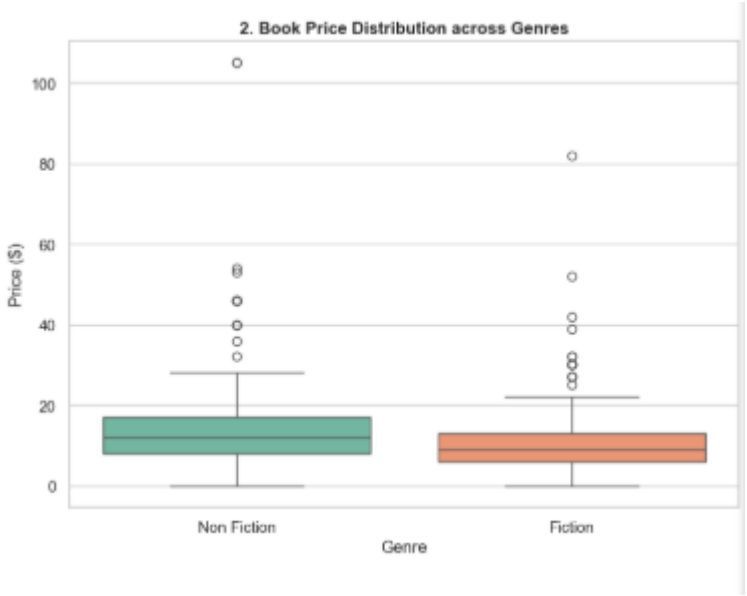
Important Inter-Feature Patterns: There is a significant positive correlation (+0.49) between Reviews and Year, indicating that books published closer to 2022 have more reviews. Furthermore, because different genres have different naming standards, Title_Length has a moderate correlation with Genre_Encoded (+0.41).

Modeling Readiness: There is no significant multicollinearity in the dataset because the majority of cross-feature correlations are still low (between -0.2 and +0.3). This ensures that **Logistic Regression** will produce stable, reliable, and highly interpretable feature coefficients and odds ratios without variance inflation distortion.



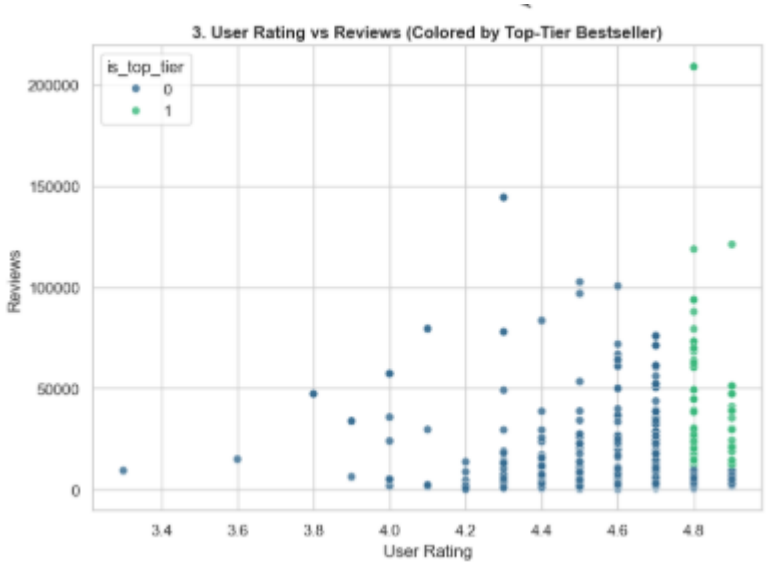
Price Distribution by Genre

The comparison box plot shows the fundamental variations in pricing methods between fiction and nonfiction bestsellers. Non-fiction books have a somewhat higher median price and a broader interquartile range (IQR), reflecting the various formats such as memoirs, self-help guides, and bulky hardcovers. While the majority of bestsellers in all categories fall below the $20 mark, Non-Fiction contains more severe high-end price outliers that exceed $40 and reach up to $105. In comparison, fiction pricing is more tightly constrained, with top outliers rarely surpassing $80. For a Logistic Regression model, this suggests that price could serve as a useful separating boundary, as higher-priced books skew heavily toward specific non-fiction subcategories.



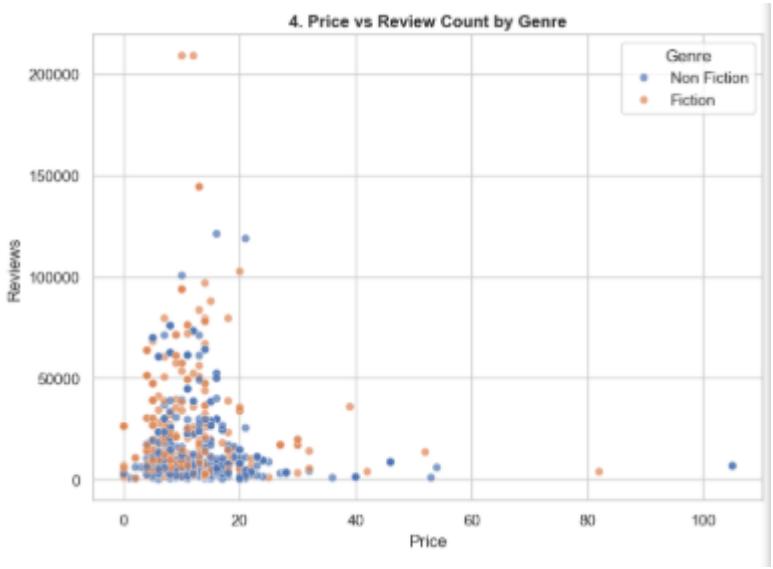
User Rating vs Reviews Scatter Plot coloured by Target

This scatterplot depicts the dual-criteria rule used to generate the target variable (is_top_tier). Top-tier books, shown in green, are on the upper right: they have ratings of 4.8 or 4.9 and review counts that exceed the dataset's median. Because the target is directly determined by these two variables, they cannot be utilised as predictors, as the model would simply be given the rule that generates the answer. They were simply utilised to define and describe the goal; the model was trained on the independent features (price, year, genre, and title_length).



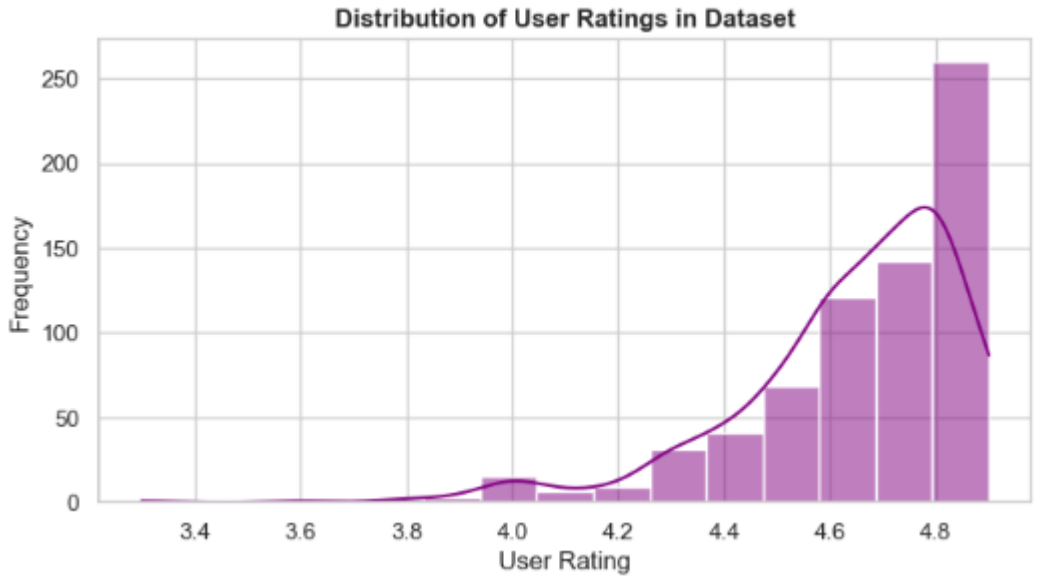
Price vs Reviews by genre Scatter Plot

This figure, which compares price to review counts across both genres, indicates an inverse link between product pricing and user interaction. The great majority of blockbuster books—those with tens or hundreds of thousands of reviews—are tightly packed at price points below $20. As book prices soar above $30, review counts plummet dramatically. This means that accessible, low-friction price is crucial in driving the high-volume readership required for a book to become a success and maintain widespread notice on Amazon.



Distribution of User Ratings in Dataset

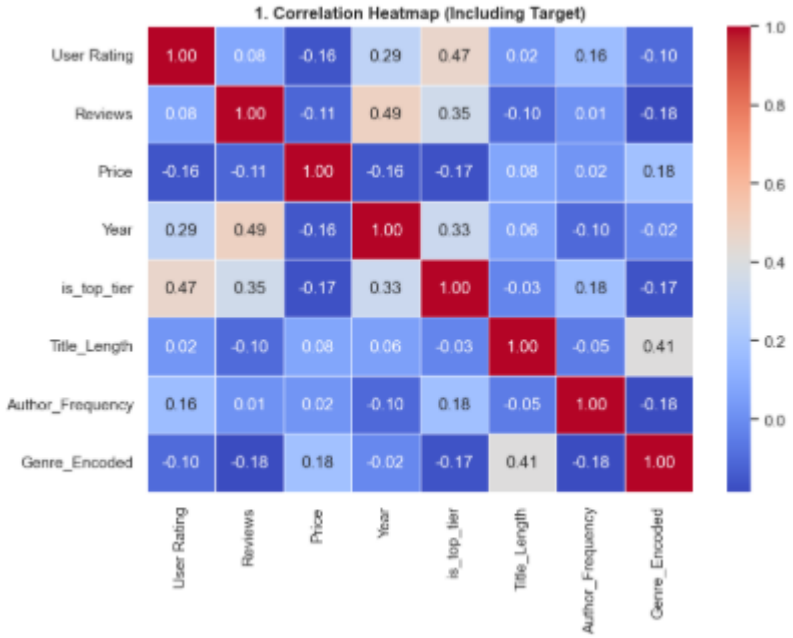
The Kernel Density Estimation (KDE) curve and histogram show a clear left-skew, with ratings primarily concentrated between 4.6 and 4.9 (peaking dramatically near 4.8). Because all records represent past Amazon bestsellers, low rated books are essentially nonexistent, creating a high baseline of customer happiness. This distribution shows the dataset's intrinsic survivorship bias. Because of the limited range in ratings, even minor decimal differences in user scores have a substantial predictive weight in predictive modelling, making them powerful features for differentiating between top-tier performances and regular blockbusters.



VISUALISATIONS WITH THE TARGET FEATURE (IS_TOP_TIER)

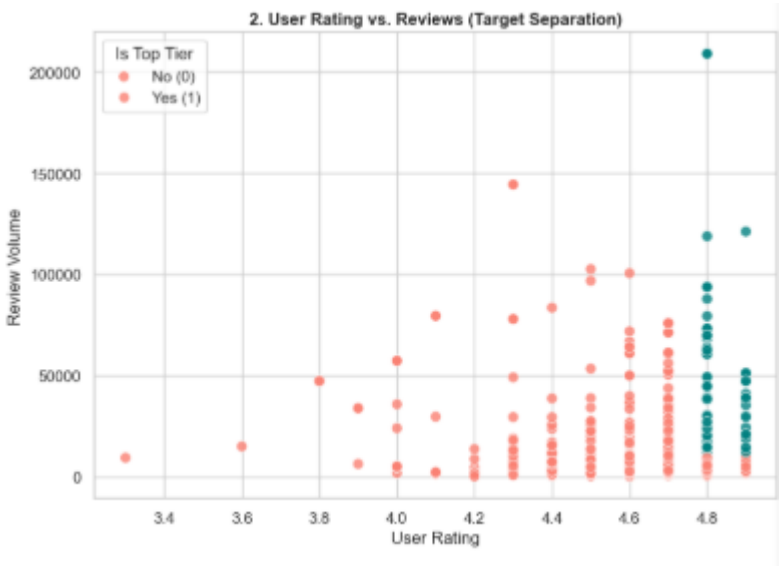
Correlation Heatmap (with Target)

This matrix uses Pearson correlation coefficients to map the linear relationships between dataset features. User Rating has the largest positive association with the goal variable is_top_tier (+0.47), followed by Reviews (+0.35) and Year (+0.33), suggesting that excellent ratings and high engagement volume are the primary drivers of elite status. Most importantly, because the bulk of cross-feature independent correlations remain low (between -0.2 and +0.3), the dataset is free of severe multicollinearity. This assures that a Logistic Regression model generates consistent, dependable, and trustworthy feature coefficients and odds ratios with no variance inflation distortion.



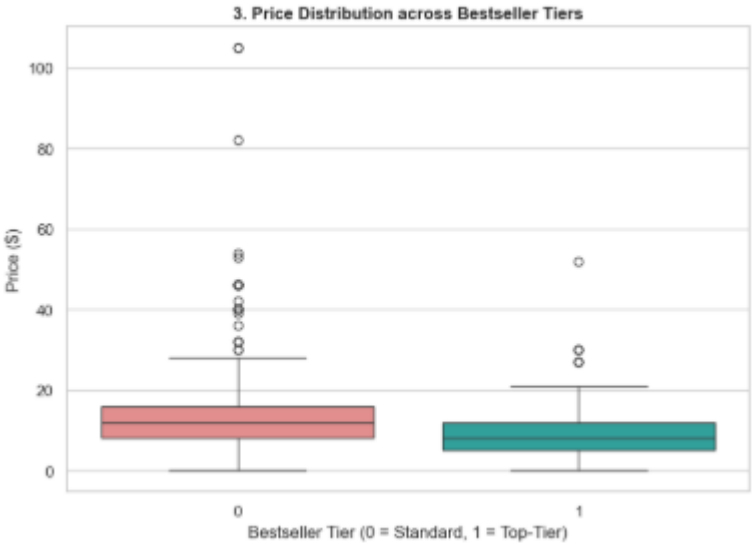
User Rating against Reviews (Target Separation)

This scatter plot repeats the point above: top-tier books (teal) are limited to ratings of 4.8 or 4.9 and review counts above the median. The clean separation is a result of how the target was defined and is not evidence of predictive power. Keeping these two variables out of the model means the results show whether Price, Year and Genre can predict top-tier status, which is a harder but more honest task.



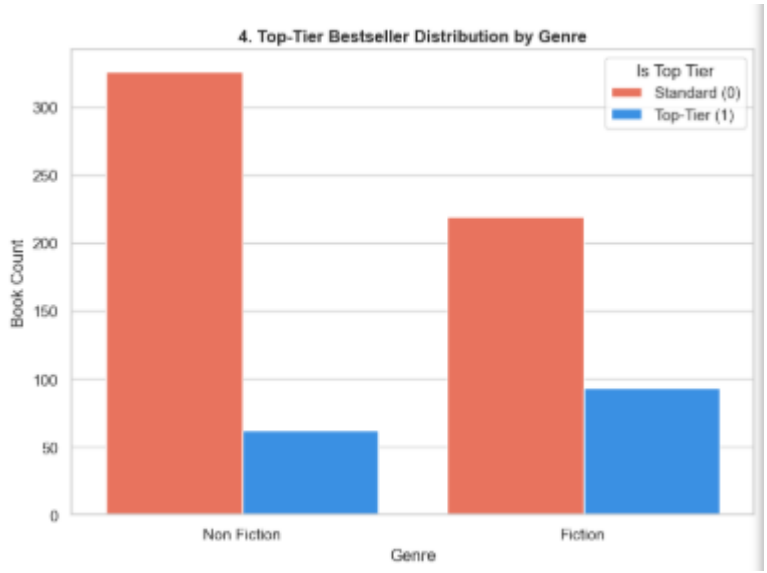
Price Distribution for Bestseller Tiers

The comparison box plot shows that Standard Bestsellers (Tier 0) have a somewhat higher median price, a broader interquartile range, and extreme outliers that exceed $100, but Top-Tier Bestsellers (Tier 1) are densely compressed at lower price points and have a lower median. This suggests that more affordable publications have a larger, more diverse audience reach, which is essential for top-tier engagement. In the context of Logistic Regression, this negative distributional shift means that pricing will likely generate a negative coefficient, meaning that lower price points enhance the log-odds of a book being a top-tier bestseller.



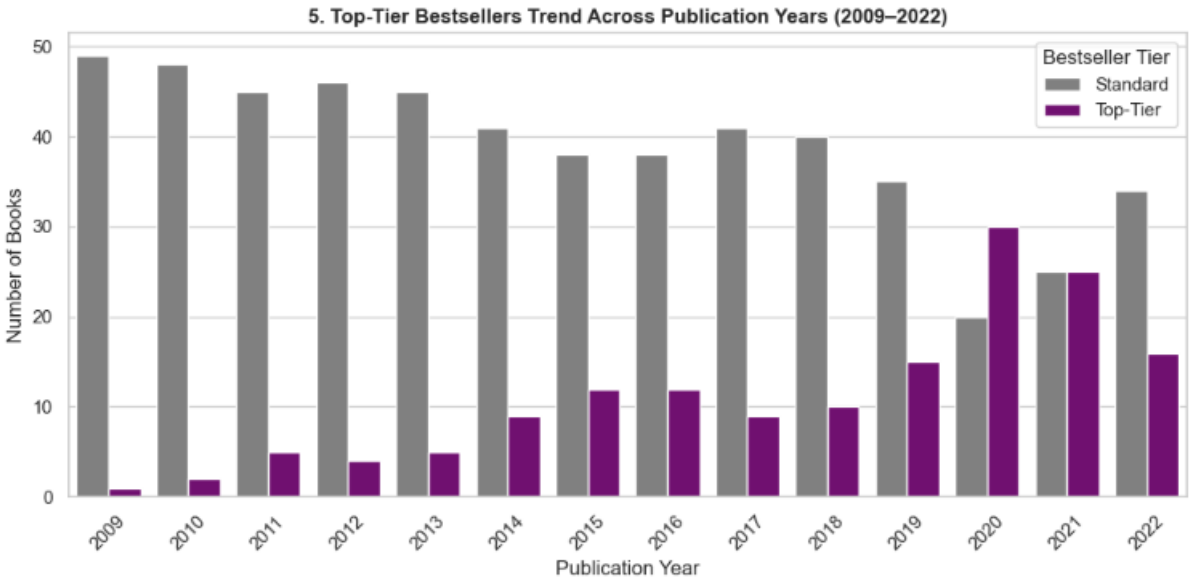
Top-Tier Bestseller Distribution By Genre

This stacked bar chart divides bestseller classifications into Nonfiction and Fiction categories. While Non-Fiction has a bigger total number of records in the dataset, Fiction has a better proportional concentration and volume of top-tier performers relative to its overall size.



Top-Tier Bestseller Trends Across Publication Years (2009–2022)

This temporal bar chart compares the annual distribution of standard versus top-tier bestsellers from 2009 to 2022. It shows a strong upward trend in the concentration of top-tier books (purple bars) in subsequent years, with peaks around 2020 and 2021, which corresponds to the +0.33 correlation between publishing year and elite rank. For Logistic Regression, this temporal drift suggests that the Year feature will be a major predictor, capturing platform growth and evolving review dynamics on Amazon over time.



Outliers

Noticeable statistical outliers were present throughout the visualisations, such as extreme book prices ranging from $30 to $105 in the pricing box plots and blockbuster titles gathering large review volumes over 150,000 in the scatter plot. These outliers were chosen specifically because they represent genuine, real-world publishing phenomena—such as expensive hardcovers, box sets, and viral literary hits—rather than data entry errors or parser artefacts. Preserving them is critical for a Logistic Regression model because they capture the genuine upper tail of commercial success and high-engagement behaviour, preventing fake data truncation and guaranteeing that the model trains on an unskewed, representative distribution of Amazon bestselling sales.

Key Insights

- **Drivers of top-tier status in the data:** User Rating (+0.47) and Reviews (+0.35) have the strongest correlations with is_top_tier, but they define the target, so they were excluded from the model.
- **Year:** the share of top-tier books grows over time (+0.33 correlation with the target), peaking in 2020 and 2021. Year became the strongest predictor in the model.
- **Pricing and accessibility:** lower prices are associated with top-tier status. Standard bestsellers have the highest prices, with outliers up to $105.
- **Genre:** Fiction has a higher share of top-tier books (about 30%) than Non Fiction (about 16%).
- **Modelling readiness:** cross-feature correlations are mostly low (between −0.2 and +0.3), so there is no severe multicollinearity among the predictors.

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b. Feature Selection & Preprocessing

Train-Test Split

Because is_top_tier is specified by User Rating and Reviews, utilising these as predictors would allow the model to glimpse the solution (target leakage). They were removed, and the candidate predictors were Price, Year, Title_Length, and Genre_Encoded.

To prevent repeated records of the same book from showing in both the training and test sets, the data was separated by book title (GroupShuffleSplit, 80/20, random_state = 42) prior to feature selection. This produced 558 training records (432 standard, 126 top-tier) and 142 test records (113 standard, 29 top-tier), with no shared titles.

Feature selection

Feature selection was then limited to the training set. The multivariate Logit model from statsmodels was fitted with Maximum Likelihood Estimation, and each feature was rated based on its p-value at $\alpha = 0.05$. Year ($p < 0.001$), Price ($p = 0.003$), and Genre_Encoded ($p = 0.013$) showed statistical significance and were maintained. Title_Length ($p = 0.283$) was not statistically significant and was dropped. The model converged normally (Pseudo $R^2 = 0.144$). This low result indicates that these predictors account for only a small portion of what makes a book top-tier, as expected when the variables that define the label are eliminated.

Feature	**Coefficient**	**p-value**	**Decision**
Price	−0.0570	0.003	Keep
Year	0.2202	< 0.001	Keep
Title_Length	0.0038	0.283	Drop
Genre_Encoded	−0.6225	0.013	Keep

The signs agree with the EDA: newer books and cheaper books are more likely to be top-tier, and Fiction (coded 0) is more likely to be top-tier than Non Fiction (coded 1).

```
Training-only Logit feature selection completed.
                           Logit Regression Results
==============================================================================
Dep. Variable:                      y   No. Observations:                  558
Model:                          Logit   Df Residuals:                      553
Method:                           MLE   Df Model:                            4
Date:                Wed, 21 Oct 2026   Pseudo R-squ.:                  0.1440
Time:                        09:45:27   Log-Likelihood:                -255.14
converged:                       True   LL-Null:                       -298.06
Covariance Type:            nonrobust   LLR p-value:                 1.008e-17
==============================================================================
                 coef    std err          z      P>|z|      [0.025      0.975]
------------------------------------------------------------------------------
const       -444.4740     62.730     -7.086      0.000    -567.422    -321.526
Price         -0.0570      0.019     -3.006      0.003      -0.094      -0.020
Year           0.2202      0.031      7.079      0.000       0.159       0.281
Title_Length   0.0038      0.004      1.073      0.283      -0.003       0.011
Genre_Encoded -0.6225      0.251     -2.485      0.013      -1.114      -0.131
==============================================================================
\nTraining-only p-values:
Year            1.455599e-12
Price           2.648878e-03
Genre_Encoded   1.297081e-02
Title_Length    2.833291e-01
dtype: float64
\nSelected independent predictors: ['Price', 'Year', 'Genre_Encoded']
Training rows: 558 | Test rows: 142
Shared title groups: 0
Train class counts: {0: 432, 1: 126}
Test class counts: {0: 113, 1: 29}
```

Scaling

The data was divided 80/20 by book title, yielding 558 training and 142 test rows (see Feature Selection & Preprocessing). Because the predictors are on quite different scales, Z-score standardisation was used (centring each characteristic at a mean of 0 with a standard deviation of 1): Year is around 2,000, price goes from $0 to $105, and genre is only 0 or 1. Putting them on the same unitless scale prevents features with large raw values from dominating gradient descent and promotes convergence.

The averages and standard deviations were determined for the training set alone (Price: mean 12.65, SD 8.92; Year: mean 2015.47, SD 3.98; Genre_Encoded: mean 0.57, SD 0.49) and then applied to both the training and test sets. This avoids test set information from leaking into training, allowing the learned coefficients to be directly compared.

```
Training Data Shape: (558, 3)
Testing Data Shape: (142, 3)
Scaling means learned from training data only: [1.26469534e+01 2.01547312e+03 5.73476703e-01]
Scaling standard deviations learned from training data only: [8.9185331  3.98498454 0.49457171]
```

c. Build and Train Logistic Regression

The Logistic Regression algorithm (Hosmer et al., 2013) was created in Python using NumPy as the LogisticRegressionScratch class and trained using batch gradient descent .

Hyperparameters: learning rate = 0.1 (a high enough step size to ensure speedy convergence without overshooting), iterations = 2,000, starting weights = 0, initial bias = 0, decision threshold = 0.5.

Training steps, repeated 2,000 times:

1. **Forward pass:** compute z = X·w + b for every training row.
2. **Sigmoid:** convert z to a probability between 0 and 1 using 1 / (1 + e^−z). Values are clipped to avoid numerical overflow.
3. **Gradients:** calculate how the loss changes with each weight and the bias: dw = (1/n)·X^T(ŷ − y) and db = (1/n)·Σ(ŷ − y).
4. **Parameter update:** move the weights and bias a small step against the gradient: w = w − lr·dw and b = b − lr·db.
5. **Loss:** record the Binary Cross-Entropy, −mean[y·log(ŷ) + (1 − y)·log(1 − ŷ)], to track learning. A tiny constant (1e-15) prevents log(0).
6. **Prediction:** after training, a probability of 0.5 or more is classified as top-tier (1), otherwise as standard (0).

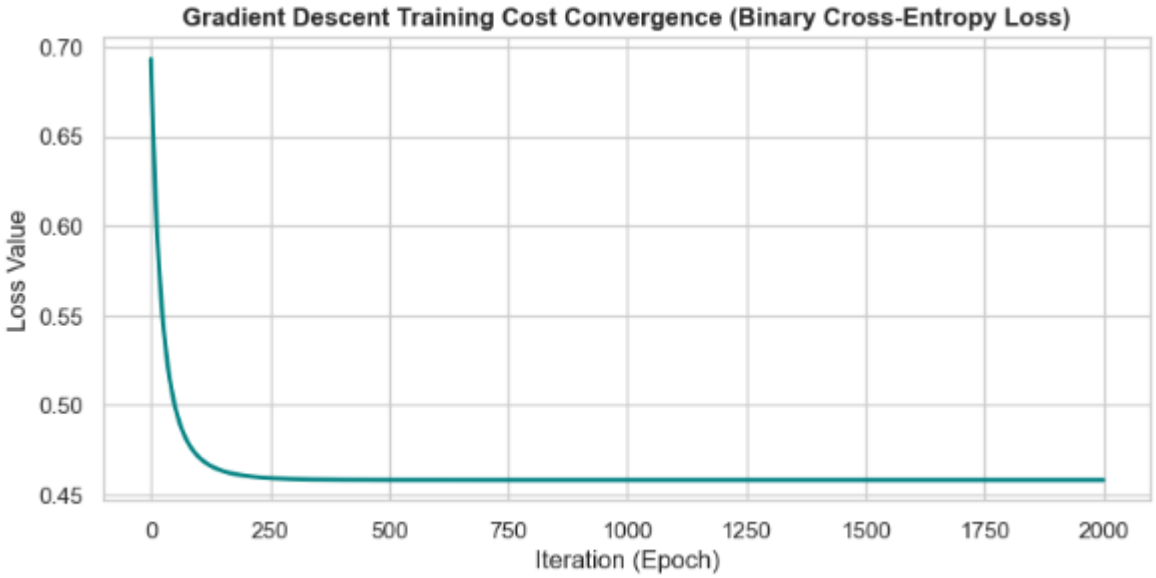
Convergence: the training loss fell from 0.6931 at the start to 0.4583 at iteration 500, and remained constant at iterations 1,000, 1,500, and 2,000. The curve flattens out after about 500 iterations, indicating that the model converged and learning was stable. Because the remaining predictors provide far less information on the label, the ultimate loss is greater than that of a model that includes ratings and reviews.

Final parameters (on standardised data): weight for Price = -0.507, weight for Year = 0.886, weight for Genre_Encoded = -0.249, bias = -1.544. The year has the greatest impact: each one-standard deviation rise (approximately four years) increases the likelihood of being in the top tier by around 2.4. Each increase of one standard

deviation in price (approximately $9) reduces the probabilities by around 40%, and nonfiction books have lower odds than fiction books.

```
Starting Training | Samples: 558, Features: 3, Iterations: 2000, Learning Rate: 0.1
Iteration 1/2000 | Loss (Binary Cross-Entropy): 0.69315
Iteration 500/2000 | Loss (Binary Cross-Entropy): 0.45833
Iteration 1000/2000 | Loss (Binary Cross-Entropy): 0.45828
Iteration 1500/2000 | Loss (Binary Cross-Entropy): 0.45828
Iteration 2000/2000 | Loss (Binary Cross-Entropy): 0.45828
Training Complete!

================================================
FINAL TRAINED MODEL PARAMETERS:
================================================
Trained Weights : [-0.5071065   0.88567044 -0.24893766]
Trained Bias    : -1.5444
================================================
```



d. Comprehensive Model Performance Analysis/ evaluation

Metrics chosen and why

- Accuracy: the total percentage of correct forecasts. When classes are uneven, it can be misleading because constantly forecasting "standard" would already result in a 79.6% exam score (113 of 142 books) (He & Garcia, 2009).
- Precision: How many of the books anticipated to be top-tier are actually such. It is important if the press wishes to avoid promoting the incorrect books.

- Recall the number of true top-tier books discovered. It counts if the press does not want to neglect good novels.
- F1-score: combines precision and recall in one number.
- ROC-AUC: how well the model outperforms standard books, regardless of the threshold utilised.
- Confusion matrix and ROC curve: visibly display the types of errors and trade-offs.

Metric and classification results:

Overall accuracy vs. class imbalance: the model achieves an accuracy of 80.28%, however the test set has a 113/29 class split, therefore predicting "standard" would result in a score of 79.6%. The model outperforms the baseline by less than a percentage point. Its accuracy stems primarily from recognising standard literature (Class 0 recall of 93%), concealing poorer performance in the minority class.

Class performance and recall trade-off: the algorithm overlooks more than two-thirds of the best books (Class 1 recall of 31%). The default 0.5 threshold treats both classes the same, regardless of the imbalance, therefore the model favours the majority class. When it does forecast top-tier, it is approximately 53% accurate.

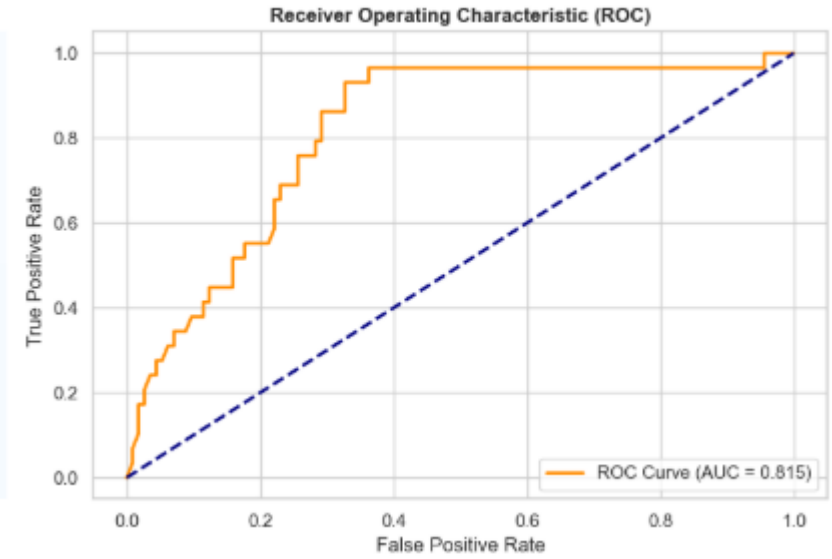
Model V1 results on the 142 test books

Metric	**Result**
Accuracy	**80.28% (114 of 142 correct)**
Precision	**0.5294**
Recall	**0.3103**
F1-score	**0.3913**
ROC-AUC	**0.8151**

Class	**Precision**	**Recall**	**F1-score**	**Support**
0 (Standard)	**0.84**	**0.93**	**0.88**	**113**
1 (Top-tier)	**0.53**	**0.31**	**0.39**	**29**
Macro average	**0.68**	**0.62**	**0.64**	**142**

Weighted average	**0.78**	**0.80**	**0.78**	**142**

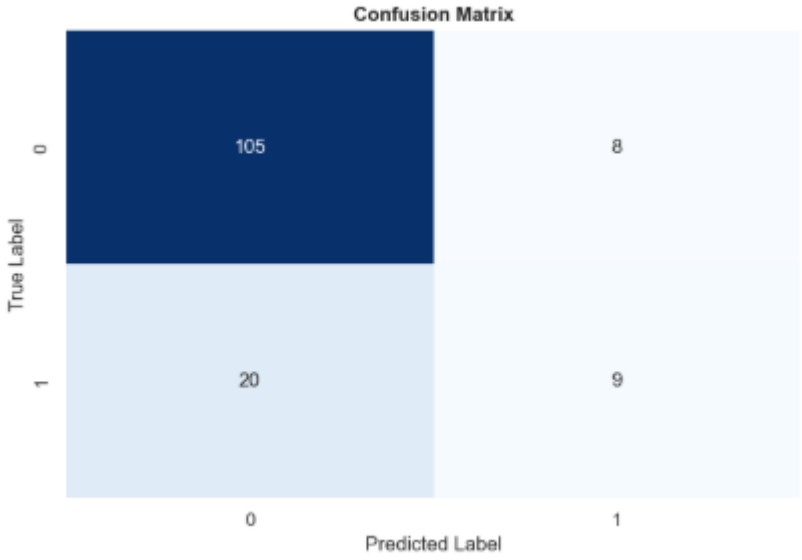
Ranking power (ROC-AUC): the ROC-AUC of 0.82 means that if one top-tier and one standard book are picked at random, the model gives the top-tier book the higher probability about 82% of the time. The ROC curve reaches a true positive rate above 0.9 at a false positive rate of about 0.3, so a lower threshold could find most top-tier books at the cost of more false alarms. The low recall is therefore largely caused by the threshold and the class imbalance, not by a complete lack of signal.



Confusion matrix and prediction vs actual results

Confusion Matrix

On the 142-book test set, the model correctly identified 105 standard books (true negatives) and 9 top-tier books (true positives). It wrongly flagged 8 standard books as top-tier (false positives) and missed 20 top-tier books (false negatives). This gives a specificity of about 93% but a sensitivity of only 31% for the top-tier class.



Comparison Table: Prediction vs. Actual

The table displays the first 15 test examples, together with their prices, years, genre codes, predicted probabilities, predicted classes, and actual classes. All 15 books were published in 2009 or 2010, had very low projected probabilities (0.004 to 0.081), and were appropriately classed as standard. This is consistent with Year being the strongest predictor: early books are rarely top-tier. Across the entire test set, 114 predictions were correct and 28 were incorrect, yielding an accuracy of 80.28%, which corresponds to the statistic above.

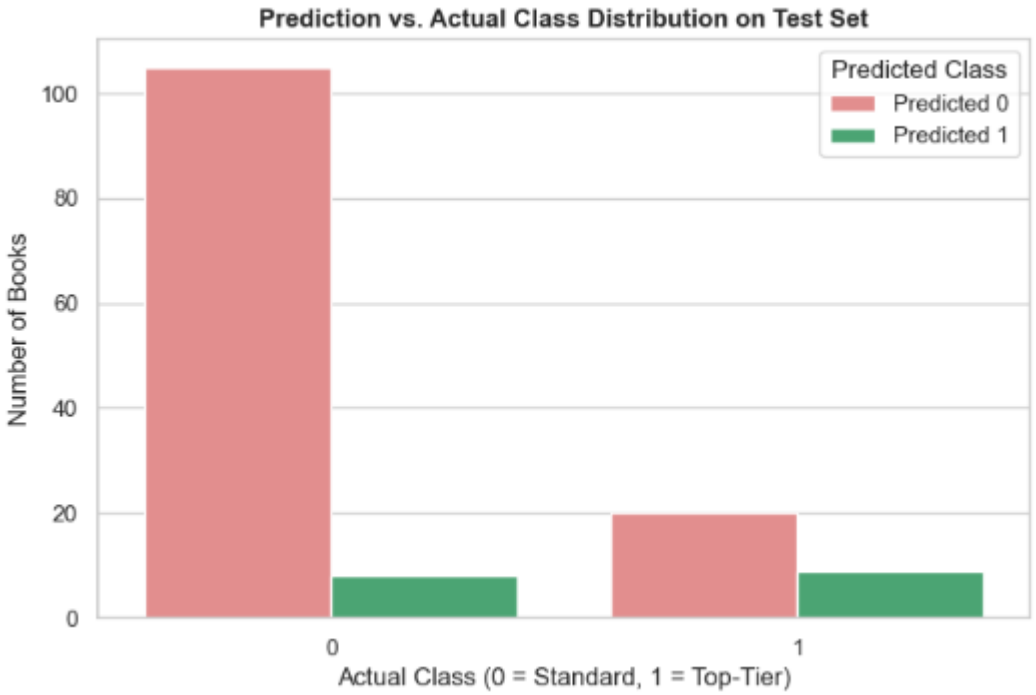
```
======================================================================
PREDICTION VS. ACTUAL COMPARISON TABLE (First 15 Test Samples)
======================================================================
```

	Price	Year	Genre_Encoded	Predicted Probability	Predicted Class (Y_pred)	Actual Class (Y_true)	Match (Correct?)
0	17.0	2009.0	1.0	0.0309	0	0	True
1	13.0	2009.0	0.0	0.0621	0	0	True
2	14.0	2009.0	1.0	0.0364	0	0	True
3	15.0	2009.0	0.0	0.0558	0	0	True
4	6.0	2009.0	1.0	0.0563	0	0	True
5	12.0	2009.0	0.0	0.0655	0	0	True
6	14.0	2009.0	1.0	0.0364	0	0	True
7	14.0	2009.0	1.0	0.0364	0	0	True
8	12.0	2009.0	1.0	0.0407	0	0	True
9	53.0	2009.0	1.0	0.0041	0	0	True
10	11.0	2009.0	0.0	0.0691	0	0	True
11	6.0	2009.0	1.0	0.0563	0	0	True
12	8.0	2009.0	0.0	0.0809	0	0	True
13	13.0	2009.0	0.0	0.0621	0	0	True
14	14.0	2010.0	1.0	0.0451	0	0	True

```
==================================================
PREDICTION VS. ACTUAL PERFORMANCE SUMMARY
==================================================
Total Test Samples Evaluated : 142
Correct Predictions (Match)  : 114
Incorrect Predictions (Miss) : 28
Actual vs Predicted Accuracy : 80.28%
==================================================
```

Prediction versus Actual Class Distribution on Test Set

The grouped bar chart shows that the model's predictions follow the majority class. Of the 29 actual top-tier books, only 9 were predicted as top-tier and 20 were predicted as standard, which visually confirms the conservative bias created by the 0.5 threshold.



Model Performance Assessment

The model performed modestly. Its accuracy of 80.28% is only little higher than the 79.6% baseline, but its ROC-AUC of 0.82 demonstrates that Price, Year, and Genre provide actual information about which books become top-tier. Standard books were reliably detected (93% recall), while the top-tier class was held back by the class imbalance and the default 0.5 threshold, resulting in a recall of 31%. These findings are lower than those of a model based on ratings and reviews, but they are more reliable because the model was not given the factors that determine the label and was tested on titles it had never seen before.

Conclusion

The exploratory data analysis and logistic regression pipeline showed that top-tier Amazon bestsellers can be partly predicted from book details that are independent of the label. By creating a binary target (is_top_tier) from user ratings and review volume, and then excluding those two variables from the predictors, the analysis avoided target leakage. A title-based split ensured that the model was tested on books it had not seen. The model found that newer books, cheaper books and

Fiction books are more likely to be top-tier, with Year as the strongest predictor. Its ROC-AUC of 0.82 shows useful ranking ability, but the default 0.5 threshold produced a conservative model that found only 31% of top-tier books.

Limitations

The study had a few methodological and structural constraints:

- The label was engineered. The top-tier designation is derived from rating and review thresholds, not sales data, therefore the model predicts a definition of success rather than actual sales.
- Few predictors. After removing the label-defining elements, only Price, Year, and Genre remained. The Pseudo R^2 of 0.144 indicates that they explain just a little portion of the result.
- Year as a proxy. Older books have had less opportunity to accumulate reviews, thus Year may represent review accumulation and Amazon's platform growth (changes in review regulations and verified-purchase badges), rather than book quality.
- Small test set. Only 29 top-tier books were in the test set, so each book changes the recall by about 3.4 percentage points, and a single hold-out split may not be representative.
- Single training run. The model was trained once, with the planned hyperparameters, and was not retrained or tuned.
- Missing metadata. Publisher marketing spend, author following, sub-genre and the full text of the books were not available.

Recommendations for future work

To build on these findings, future research should adjust the probability threshold, such as using Youden's J statistic or a precision-recall curve, to increase minority-class recall (scikit-learn, n.d.), and class weights or resampling to address the imbalance. Non-linear models like Random Forest or XGBoost have the potential to capture more intricate correlations between price, year, and genre (IBM, 2024). Richer attributes, such as publisher, page count, series information, sentiment from

review text, and author history, would provide the model with additional data to learn from. Finally, a real outcome such as sales rank should replace the engineered label, and grouped cross-validation by title should be used to make the results more stable.

References

IBM. (2024, May 09). Retrieved October 21, 2026, from What is XGBoost?: https://www.ibm.com/think/topics/xgboost

scikit-learn. (n.d.). Retrieved October 21, 2026, from Tuning the decision threshold for class prediction: https://scikit-learn.org/stable/modules/classification_threshold.html